

Eigenvalues of tetrahedral sums of Hermitian matrices

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Abstract

This is the abstract.

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Outline Draft May 30 2024

Below is the proposed outline of the paper as of May 30, 2024. Our proposed journal is *Communications in Mathematical Physics*. It will be ideal to illustrate with figures.

- introduction (include 2x2 case)
- general definition
- relationship to Horn's problem, (how does it generalize the problem)
- representation theoretic/quantization analogue (Ponzano-Regge and connection to state-sum models? etc.)
- comment about deviation from Thompson's conjecture (correspondence between eigenvalues of sums and logs of singular values of products)
- spectral embedding into the quantum marginal problem
- derivation of entropic inequalities (both from rep theory directly, and from embedding into QMP, things like concavity and data processing inequality for Renyi-entropies, etc), then prove they go beyond face constraints

1 Introduction

Notational conventions: Throughout this paper, $d \in \mathbb{N}_{>0}$ will denote the size of the matrices under consideration. We also exclusively work with matrices with coefficients in the field of complex numbers $\mathbb{C}$, so for notational brevity we write $\mathrm{GL}(d)$ as shorthand for $\mathrm{GL}(d, \mathbb{C})$. Weights for $\mathrm{GL}(d)$ will be denoted using Greek letters and will be identified with d -tuples of integers, e.g. $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{Z}^d$. The weight lattice for $\mathrm{GL}(d)$ is denoted by $\Lambda^{\mathbb{Z}}(d) \simeq \mathbb{Z}^d$ in the usual way; a weight $\alpha \in \Lambda^{\mathbb{Z}}(d)$ is dominant if $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_d$ and the set of all dominant weights for $\mathrm{GL}(d)$ is denoted by $\Lambda_{\geq}^{\mathbb{Z}}(d)$.

Irreducible highest weight representations of $\mathrm{GL}(d)$ with weight $\lambda \in \Lambda^{\mathbb{Z}}(d)$ are denoted by L_{λ} and irreducible representations of S_n of highest weight $\lambda \in \Upsilon_n$ are denoted by V_{λ} .

The set of all $d \times d$ complex Hermitian matrices will be denoted by $\mathcal{H}(d)$ while the set of all possible eigenvalues of a $d \times d$ complex Hermitian matrix is denoted by $\Lambda_{\geq}^{\mathbb{R}}(d) \subset \mathbb{R}^d$. The set of $d \times d$ Hermitian matrices X with eigenvalues $\mathrm{eig}(X) = (x_1, \dots, x_d) \in \Lambda_{\geq}^{\mathbb{R}}(d)$ will be denoted by $\mathcal{O}_x$.

2 Generalizations of Horn's problem

This section will include a formal statement of the problem under investigation, its relation to Horn's original problem and other generalizations of Horn's problem. This section will hopefully briefly describe what is known about the eigenvalues of triangular sums of matrices. Regarding tetrahedral spectra, we will describe many of the symmetries of the problem statement and various "gauge" choices that one could make. Additional generalizations of the problem statement, like to more matrices, linear combinations, etc. will be mentioned here. It will be important to mention the manner in which the analogue of the "Thompson-Klyachko" theorem does not hold, i.e. singular values of tetrahedral products are not the exponentials of eigenvalues of tetrahedral sums in general.

2.1 Graphical Horn problems

Definition 1. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$ together with source and target maps $s, t : \mathcal{E} \rightarrow \mathcal{V}$. Let d be a positive integer. A *spectral map* (for $\mathcal{G}$ of dimension d) is a function $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ assigning to each edge $e \in \mathcal{E}$ in the directed graph a d -tuple of real numbers $\lambda(e) = (\lambda_1(e), \dots, \lambda_d(e))$. The map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is said to be *realizable (by Hermitian matrices)* if there exists a map $H : \mathcal{E} \rightarrow \mathcal{H}(d)$ assigning to each edge $e \in \mathcal{E}$ a $d \times d$ Hermitian matrix $H(e)$ such that

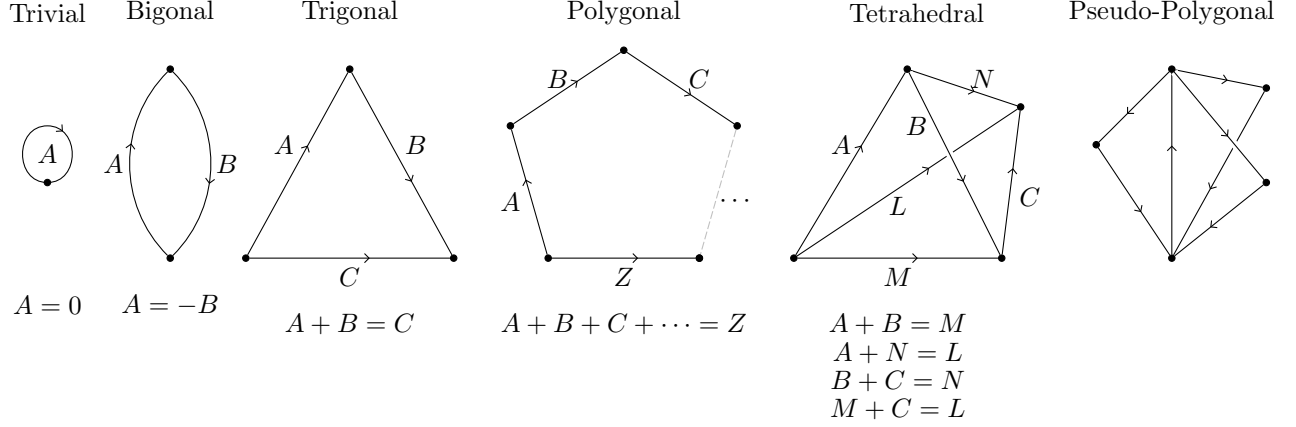


Figure 1: Examples of graphical Horn problems.

- for each edge $e \in \mathcal{E}$, $H(e)$ has eigenvalues $\lambda(e)$,

$$\text{eig}(H(e)) = \lambda(e), \quad \text{\texttt{eq:spec_cond}} \quad (1)$$

- for each cycle $\ell \subseteq \mathcal{E}$ with orientation map $\gamma_\ell : \ell \rightarrow \{\pm 1\}$,

$$\sum_{e \in \ell} \gamma_\ell(e) H(e) = 0. \quad \text{\texttt{eq:cycle_cond}} \quad (2)$$

Alternatively, one can equivalently encode the constraints imposed by Eq. (2) in terms of a $\{-1, 0, +1\}$ -valued $L \times |E|$ matrix (where L is the number of (generating) cycles in $\mathcal{G}$), denoted by $\Gamma(\mathcal{G})$, where

$$\Gamma(\mathcal{G})_{\ell, e} = \begin{cases} 0 & e \notin \ell, \\ \gamma_\ell(e) & e \in \ell. \end{cases} \quad (3)$$

In terms of $\Gamma(\mathcal{G})$, Eq. (2) reads for all cycles ℓ :

$$\sum_{e \in \mathcal{E}} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0 \quad (4)$$

\texttt{prob:ghp}

Problem 1 (Graphical Horn problem). Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph, let d be a positive integer. The *graphical Horn problem* is to characterize the set of realizable spectral maps $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ for $\mathcal{G}$ of dimension d .

There might be an important connection to the their of quivers and representations of quivers. In particular, the notion of a morphism between a representation of a quiver in terms of operators on vertices forming computing squares models precisely the kinds of continuous symmetries which may be present in a given graphical Horn problem. Indeed, the spectral data were are ultimately interested in is equivalent to the orbit under such “local” isomorphisms (i.e. conjugation by isomorphisms).

TC: It turns out that the only relavent data for statement of a graphical Horn problem is the matrix $\Gamma(\mathcal{G})$; perhaps its best to drop reference to $\mathcal{G}$ altogether and just have the problem statement depend on a matrix Γ ? Moreover, one could in principle take the matrix to be $\mathbb{R}$ -valued instead of $\{-1, 0, +1\}$ -valued, although that would make the connection to the representation theory even weaker. Generalizing to $\mathbb{Z}$ -valued Γ is also essentially the same I think...

There are a few operations which preserve the set of realizable spectral maps.

Lemma 1. Consider a graphical Horn problem of dimension d .

- The zero spectral map $0_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ assigning to $e \in \mathcal{E}$, the spectrum $0_d(e) = (0, \dots, 0)$ is realizable.

- If $\alpha : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(a)$ and $\beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(b)$ are realizable, then their direct sum $\alpha \oplus \beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(a+b)$ and tensor product $\alpha \otimes \beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(ab)$ (defined pointwise in the obvious manner), are both realizable.
- If $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable and $p \in \mathbb{R}_{>0}$ a positive real number, then $p\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$, defined by $(p\lambda)(e) = (p\lambda_1(e), \dots, p\lambda_d(e))$ is also a realizable.
- If $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable, then $\lambda^* : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$, defined by a pointwise application of the negation-reversal involution, is also realizable.

Remark 1. Graphical Horn problems can be reframed as a question about the simultaneous vanishing of a moment map in the sense of symplectic geometry. Throughout this paper, we identify the set of $d \times d$ Hermitian matrices H , $\mathcal{H}(d)$ with the set $\mathfrak{u}(d)^*$ of linear functionals on the Lie algebra $\mathfrak{u}(d)$ of $U(d)$ by the pairing $(H, X) = i \operatorname{Tr}(XH)$. Given a spectral map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ and a cycle ℓ , we consider the symplectic manifold

$$\mathcal{M}_{\ell} := \prod_{e \in \mathcal{E}} \mathcal{O}_{\lambda(e)\gamma_{\ell}(e)} \quad (5)$$

where $\lambda(e)^{+1} = \lambda(e)$ and $\lambda(e) = \lambda(e)^*$ and where $\mathcal{O}_{\mu}$ is the set of Hermitian $d \times d$ matrices with eigenvalues $\mu_1 \geq \dots \geq \mu_d$ which may be identified with a corresponding coadjoint orbit in $\mathfrak{u}(d)^*$ (after identifying μ with the diagonal $d \times d$ matrix $\operatorname{diag}(\mu_1, \dots, \mu_d)$) equipped with the standard KKS symplectic form. Then define a Hamiltonian $U(d)$ -action on $\mathcal{M}_{\ell}$ where u acts diagonally on $H(e)$ by conjugation ($u \cdot H(e) = uH(e)u^{-1}$). It can be shown that the moment map $\Phi_{\ell} : \mathcal{M}_{\ell} \rightarrow \mathcal{H}(d)$ associated to this action satisfies

$$\Phi_{\ell}(\gamma_{\ell}(e)H(e) \mid e \in \ell) = \sum_{e \in \ell} \gamma_{\ell}(e)H(e) \quad (6)$$

and thus vanishes precisely when the cycle condition, Eq. (2), is satisfied.

TODO: Show how to generalize the construction due to Roberts [Rob99] by taking the Cartesian product of these manifolds over all cycles and then imposing additional group actions and moment-map conditions to stitch everything together.

Example 1 (The trigonal Horn problem). Perhaps the smallest non-trivial example of a graphical Horn problem is the one for which Horn originally conjectured a solution [Hor62], namely, what are the possible eigenvalues, (α, β, γ) , of a triple of $d \times d$ Hermitian matrices (A, B, C) satisfying the condition $A + B = C$? As a graphical Horn problem,

$$\Gamma = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \quad (7)$$

From the perspective of Problem 1, we refer to this question as the *trigonal Horn problem*.

Example 2 (Polygonal Horn problems). If the directed graph $\mathcal{G}$ consists of a single closed cycle, then the corresponding Horn problem will be referred to as a *polygonal Horn problem*; the polygonal Horn problem associated to an *oriented* cycle $(e_1, e_2, \dots, e_r, e_1)$ consisting of r edges asks what are the possible eigenvalues $(\lambda_1, \dots, \lambda_r)$ of r Hermitian matrices $(H_1, \dots, H_r)$ with zero-sum $H_1 + \dots + H_r = 0$.

$$\Gamma = \begin{pmatrix} 1 & \dots & 1 \end{pmatrix} \quad (8)$$

Remark 2. Evidently, there is a bijection between the set of realizable spectra (α, β, γ) and the set of realizable spectra $(\lambda_1, \lambda_2, \lambda_3)$ for the polygonal Horn problem where $r = 3$: namely

$$(\lambda_1, \lambda_2, \lambda_3) = (\alpha, \beta, \gamma^*) \quad (9)$$

where $\gamma \mapsto \gamma^*$ is the involution on $\Lambda_{\geq}^{\mathbb{R}}(d)$ defined by negation of entries and then reversing to preserve the ordering:

$$\gamma^* = (-\gamma_d, \dots, -\lambda_1). \quad (10)$$

Example 3 (Tetrahedral Horn problem). In this paper, a considerable focus will be placed on the *tetrahedral Horn problem* which is the graphical Horn problem associated with the directed graph above forming a tetrahedron. Specifically, a sextuple of candidate spectra,

$$(\alpha, \beta, \gamma, \mu, \nu, \lambda) \in (\Lambda_{\geq}^{\mathbb{R}}(d))^{\times 6} \quad (11)$$

are said to be *tetrahedral* if they are the respective eigenvalues of Hermitian matrices

$$(A, B, C, M, N, L) \in \mathcal{H}(d)^{\times 6} \quad (12)$$

satisfying the following four conditions:

$$A + B = M, \quad M + C = L, \quad B + C = N, \quad A + N = L. \quad (13)$$

As a graphical Horn problem,

$$\Gamma = \begin{pmatrix} 1 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \quad (14)$$

Example 4 (One-dimensional graphical Horn problems). An important special case of graphical Horn problem for a digraph $\mathcal{G}$ is the case where $d = 1$. If $d = 1$, then vector space of 1×1 Hermitian matrices becomes isomorphic to the real numbers, $\mathcal{H}(1) \simeq \mathbb{R}$, and similarly the space of candidate eigenvalues becomes $\Lambda_{\geq}^{\mathbb{R}}(1) = \mathbb{R}$. Moreover, the spectral map $\lambda : \mathcal{E} \rightarrow \mathbb{R}$ becomes an assignment of real numbers of the edges of $\mathcal{G}$, the map $H : \mathcal{E} \rightarrow \mathbb{R}$ must necessarily equal spectral map λ by the spectral condition Eq. (1), and the cycle condition characterizing realizable spectral maps simplifies to the condition that λ , viewed as a vector $(\lambda(e))_{e \in \mathcal{E}} \in \mathbb{R}^{|\mathcal{E}|}$, must lie in the kernel of $\Gamma(\mathcal{G})$,

$$\lambda \in \ker(\Gamma(\mathcal{G})). \quad \text{\textcolor{blue}{eq:ker_cond_d_1}} \quad (15)$$

In fact, this simple condition provides a complete solution to all graphical Horn problems where $d = 1$.

The solution to all one-dimensional graphical Horn problems provided by Eq. (15) also gives rise to a simple necessary condition for $d > 1$ graphical Horn problems.

lem:trace_cond

Lemma 2. Consider the graphical Horn problem of dimension d for the digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Then the spectral map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is realizable only if

$$\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G})), \quad (16)$$

where, by abuse of notation, $\text{Tr} : \Lambda_{\geq}^{\mathbb{R}}(d) \rightarrow \mathbb{R}$ denotes the sum

$$\text{Tr}((\lambda_1, \dots, \lambda_d)) := \lambda_1 + \dots + \lambda_d. \quad (17)$$

Proof. Suppose that $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a spectral map realized by the assignment $H : \mathcal{E} \rightarrow \mathcal{H}(d)$, meaning $\text{eig}(H(e)) = \lambda(e)$ for all edges $e \in \mathcal{E}$, and $\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0$ for all cycles ℓ . Taking the trace of cycle condition Eq. (2) yields

$$0 = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(H(e)) = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(\lambda(e)). \quad (18)$$

In other words, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ which implies $\text{Tr} \circ \lambda$ is a realizable spectral map for the $d = 1$ case. *lem:ker_shift* $\square$

Lemma 3. Consider the graphical Horn problem of dimension d for the digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Suppose $\mu : \mathcal{E} \rightarrow \mathbb{R}$ lies in the kernel of $\Gamma(\mathcal{G})$ and define $\mu \cdot 1_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ to be the spectral map defined by

$$(\mu \cdot 1_d)(e) := (\mu(e), \dots, \mu(e)). \quad (19)$$

Then $\mu \cdot 1_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable spectral map. Moreover, $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable spectral map if and only if $\lambda + \mu \cdot 1_d$ is a realizable.

Proof. The spectral map $\mu \cdot 1_d$ is realized by the assignment $H_{\mu} : \mathcal{E} \rightarrow \mathcal{H}(d)$ defined by

$$H_{\mu}(e) := \mu(e)I_d \quad (20)$$

where I_d is the $d \times d$ identity matrix. This is because it was assumed that $\mu \in \ker(\Gamma(\mathcal{G}))$ and therefore

$$\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} H_\mu(e) = \left(\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \mu(e) \right) I_d = 0. \quad (21)$$

Now suppose $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is realized by the assignment $H_\lambda : \mathcal{E} \rightarrow \mathcal{H}(d)$, then it follows that $H_\lambda + H_\mu$, defined by

$$(H_\lambda + H_\mu)(e) = H_\lambda(e) + \mu(e) I_d \quad (22)$$

realizes the spectral map $\lambda + \mu \cdot 1_d$ because the spectrum of $H_\lambda(e) + \mu(e) I_d$ is simply $\lambda(e) + \mu \cdot 1_d(e)$ (the eigenvalues add because $[H_\lambda(e), I_d] = 0$). $\square$

A direct corollary of Lemma 3 is that the set of realizable spectral maps, $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$, is entirely determined by the set of realizable and *traceless* spectral maps in the sense that $\text{Tr} \circ \lambda = 0_d$.

Corollary 4. *Let $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ be a spectral map such that $\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G}))$ and let $\lambda^0 : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ be the traceless spectral map defined by*

$$\lambda^0(e) := (\lambda_1(e) - \text{Tr}(\lambda(e)), \dots, \lambda_d(e) - \text{Tr}(\lambda(e))). \quad (23)$$

Then λ is realizable if and only if λ^0 is realizable.

Proof. The spectral map λ^0 is obtained from λ by subtracting the spectral map $\text{Tr} \circ \lambda$. By assumption, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ by Lemma 2, and thus Lemma 3 (when $\mu \mapsto \pm \text{Tr} \circ \lambda$) implies λ^0 is realizable if and only if λ is realizable. $\square$

Corollary 5. *Suppose $\mathcal{G}$ is a graphical Horn problem such that $\ker(\Gamma(\mathcal{G}))$ contains a one-dimensional spectral map $\rho : \mathcal{E} \rightarrow \mathbb{R}$ which is strictly positive, meaning*

$$\forall e \in \mathcal{E} : \rho(e) > 0. \quad (24)$$

Let $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ be a spectral map. Then there exists a sufficiently large positive $p \in \mathbb{R}_{>0}$ such that $\lambda^+ := \lambda + p\rho \cdot 1_d$ is a spectral map consisting of strictly positive spectra meaning

$$\forall e \in \mathcal{E} : \lambda_d^+(e) > 0. \quad (25) \quad \text{\textcolor{blue}{(eq:positive_spectra)}}$$

Proof. The value of $\lambda_d^+(e)$ for $e \in \mathcal{E}$ is determined by

$$\lambda_d^+(e) = \lambda_d(e) + p\rho(e). \quad (26)$$

In order to satisfy Eq. (25) it suffices to let p satisfy

$$p > \max_{e \in \mathcal{E}} \frac{-\lambda_d(e)}{\rho(e)}. \quad (27)$$

$\square$

Example 5. The tetrahedral Horn problem defined above admits of strictly positive one-dimensional spectral maps. Specifically, $\rho = (a, b, c, a+b, b+c, a+b+c)$ lies in the kernel of Γ and is strictly positive whenever $a, b, c > 0$.

3 Representation theory

The purpose of this section is widen the bridge between representation theory and Horn-type problems. We will begin with a treatment of the connection between triangular spectra and the Littlewood-Richardson coefficients and discuss its connection to symplectic geometry and moment maps. Then we will articulate the manner in which tetrahedral spectra are related to the norm of the recoupling operator for $\text{SU}(d)$ and discuss the manner in which convexity results for moment maps cannot be applied and thus one must not expect the space of tetrahedral spectra to be convex (indeed it is not).

3.1 Triangular spectra & Littlewood-Richardson coefficients

Definition 2. Let $\alpha, \beta, \gamma \in \Lambda_{\geq}^{\mathbb{Z}}(d)$ be a triple of dominant weights for $\mathrm{GL}(d)$ and let $L_{\alpha}, L_{\beta}, L_{\gamma}$ be the triple of irreducible representations of $\mathrm{GL}(d)$ with corresponding highest weights. Let $C_{\alpha, \beta}^{\gamma}$ denote the multiplicity space of $\mathrm{GL}(d)$ -intertwiners from L_{γ} to $L_{\alpha} \otimes L_{\beta}$:

$$C_{\alpha, \beta}^{\gamma} := \mathrm{Hom}_{\mathrm{GL}(d)}(L_{\gamma}, L_{\alpha} \otimes L_{\beta}). \quad (28)$$

The *Littlewood-Richardson coefficient* is defined as

$$c_{\alpha, \beta}^{\gamma} := \dim(C_{\alpha, \beta}^{\gamma}). \quad (29)$$

Using the theory of spectrum estimation developed in the context of quantum information theory [ARS88; KW01; HM02], the following pair of theorems were proven in [Chr08]. Recall that $(\alpha, \beta, \gamma) \in \Lambda_{\geq}^{\mathbb{R}}(d)^{\times 3}$ are said to be *trigonal* if there exists Hermitian matrices $A, B, C \in \mathcal{H}(d)$ satisfying (i) $A + B + C = \bar{0}$ and (ii)

$$(\mathrm{eig}(A), \mathrm{eig}(B), \mathrm{eig}(C)) = (\alpha, \beta, \gamma). \quad (30)$$

Theorem 6. If $c_{\alpha, \beta, \gamma} > 0$ for some $\alpha, \beta, \gamma \in \Lambda_{\geq}^{\mathbb{Z}}(d)$, then (α, β, γ) are trigonal.

Theorem 7. If $(\alpha, \beta, \gamma) \in \Lambda_{\geq}^{\mathbb{R}}(d)^{\times 3}$ are triangular, then there exists a graded sequence $(\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)) \in \Lambda_{\geq}^{\mathbb{Z}}(d)^{\times 3}$ such that (i) for all $n \in \mathbb{N}_{>0}$,

$$c_{\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)} > 0, \quad (31)$$

and (ii)

$$\lim_{n \rightarrow \infty} \frac{\tilde{\alpha}(n)}{n} = \alpha, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\beta}(n)}{n} = \beta, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\gamma}(n)}{n} = \gamma. \quad (32)$$

3.2 Tetrahedral spectra & recoupling operators

defn:young;recouple

Definition 3. Let $a, b, c \in \mathbb{N}_{>0}$ be given and let

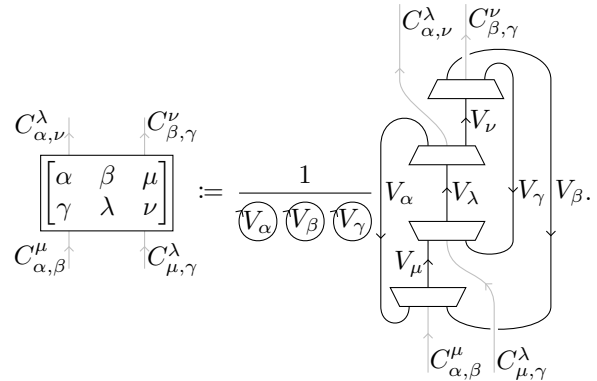
$$\alpha \in \Upsilon_a, \quad \beta \in \Upsilon_b, \quad \gamma \in \Upsilon_c, \quad \mu \in \Upsilon_{a+b}, \quad \nu \in \Upsilon_{b+c}, \quad \lambda \in \Upsilon_{a+b+c}. \quad (33)$$

The *Young recoupling operator* is the operator

$$\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} : C_{\alpha, \beta}^{\mu} \otimes C_{\mu, \gamma}^{\lambda} \rightarrow C_{\alpha, \nu}^{\lambda} \otimes C_{\beta, \gamma}^{\nu}, \quad (34)$$

between the multiplicity spaces $C_{\alpha, \beta}^{\mu} := \mathrm{Hom}_{S_a \times S_b}(V_{\alpha} \otimes V_{\beta}, V_{\mu})$ defined as

$$\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} := \frac{\mathrm{Tr}_{V_{\alpha} \otimes V_{\beta} \otimes V_{\gamma}}[(\mathrm{id}_{V_{\alpha}} \otimes Y_{\beta \gamma}^{\nu; \dagger}) Y_{\alpha \nu}^{\lambda; \dagger} Y_{\mu \gamma}^{\lambda} (Y_{\alpha \beta}^{\mu} \otimes \mathrm{id}_{V_{\gamma}})]}{\dim(V_{\alpha} \otimes V_{\beta} \otimes V_{\gamma})} \quad (35)$$



$$\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} := \frac{1}{\bigcirc V_{\alpha} \bigcirc V_{\beta} \bigcirc V_{\gamma}} \quad (36)$$

3.3 An entropic inequality for tetrahedral spectra

In Section 3.2, it was shown that the realizability of a given sextuple of candidate eigenvalues, $(\alpha, \beta, \gamma, \mu, \nu, \lambda)$, for the tetrahedral Horn problem is encoded both the asymptotics of the a recoupling operator for $\text{GL}(d)$ as well as a recoupling operator for the Young subgroups of the symmetric group. The purpose of this section is to prove an upper bound for the norm of the Young recoupling operator whose asymptotic limit gives rise to an entropic inequality which is necessarily satisfied by realizable spectra.

thm:young_recouple_bound

Theorem 8. Let $\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix}$ be the Young recoupling operator (Definition 3). Then

$$\left\| \begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} \right\|_{HS}^2 \leq \left(c_{\alpha, \beta}^{\mu} c_{\beta, \gamma}^{\nu} c_{\mu, \gamma}^{\lambda} c_{\alpha, \nu}^{\lambda} \right)^{\frac{1}{2}} \frac{\dim(V_{\nu}) \dim(V_{\mu})}{\dim(V_{\lambda}) \dim(V_{\beta})} \quad (37)$$

Proof. See Appendix A. □

TODO: Figure out how to take the limit of $\dim(V_{\lambda})$ when $\lambda \vdash pn$ as $n \rightarrow \infty$; the exact form of the entropic inequality is sensitive to how this is done.

4 The spectral quantum marginal problem

The purpose of this section will be to re-articulate the spectral embedding defined in [CŞW18] and state the sense in which the tetrahedral spectra problem is an instance of the quantum marginal problem. From here we can discuss the extent to which known inequalities coming from quantum information theory yield constraints necessarily satisfied by tetrahedral spectra.

4.1 A spectral embedding

defn:spec_embed

Definition 4. Let $(H_1, \dots, H_k) \in \mathcal{H}(d)^{\times k}$ be a k -tuple of $d \times d$ positive-semidefinite Hermitian matrices and let X be a subset of $\{1, \dots, k\}$. Let $\{|1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^d$ and $\{|0\rangle, |1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^{d+1}$. Define $H_X := \sum_{i \in X} H_i$ as the partial sum. Define a Hermitian operator, Φ_X , acting on $(\mathbb{C}^{d+1})^{\otimes k}$ with coefficients with respect to the computational bases $|i_1 \dots i_k\rangle$ be given by the following formula:¹

$$\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle = \begin{cases} \langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle & \exists! i_p, j_q > 0, p, q \in X \\ 0 & \text{otherwise.} \end{cases} \quad (38)$$

Note that, up to shuffling the $k+1$ copies of $\mathbb{C}^{d+1}$ in $(\mathbb{C}^{d+1})^{\otimes k}$, Φ_X is of the form $\tilde{\Phi}_X \otimes Z_{[k] \setminus X}$ where

$$Z_X = |0 \dots 0\rangle \langle 0 \dots 0|. \quad (39)$$

and where $\tilde{\Phi}_X$ acts on $(\mathbb{C}^{d+1})^{\otimes |X|}$. Also define the operator Γ_X which is $\tilde{\Phi}_X$ shifted by Z_X

$$\Gamma_X := \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \quad (40)$$

and in particular $\Gamma_{\emptyset} := 1$.

¹Specifically, $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ is non-zero only if there exists a unique $p \in [k]$ such that $i_p > 0$ and a unique $q \in [k]$ such that $j_q > 0$ (with all other indices $i_{p'}$ and $j_{q'}$ for $p' \neq p$ and $q' \neq q$ being zero). Furthermore, if $p, q \in [k]$ are the unique such indices, then $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ is zero if either $p \notin X$ or $q \notin X$; otherwise $p, q \in X$ and then $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ takes the value $\langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle$, the inner product between the i_p th column of $H_p^{\frac{1}{2}}$ and the j_q th column of $H_q^{\frac{1}{2}}$.

Example 6. Given two Hermitian operators $(H_1, H_2) \in \mathcal{H}(d)^{\times 2}$, the matrix $\Gamma_{1,2}$ has the following coefficients for all $i, j, k, l \in \{1, \dots, d\}$.

$$\langle 00 | \Gamma_{1,2} | 00 \rangle = 1 - \text{Tr}(H_{1,2}), \quad (41)$$

$$\langle i0 | \Gamma_{1,2} | j0 \rangle = \langle i | H_1 | j \rangle, \quad (42)$$

$$\langle 0i | \Gamma_{1,2} | 0j \rangle = \langle i | H_2 | j \rangle, \quad (43)$$

$$\langle i0 | \Gamma_{1,2} | 0j \rangle = \langle i | H_1^{\frac{1}{2}} H_2^{\frac{1}{2}} | j \rangle, \quad (44)$$

$$\langle 0i | \Gamma_{1,2} | j0 \rangle = \langle i | H_2^{\frac{1}{2}} H_1^{\frac{1}{2}} | j \rangle, \quad (45)$$

$$\langle ik | \Gamma_{1,2} | jl \rangle = 0. \quad (46)$$

lem:spec_embed

Lemma 9. Let everything be defined as in Definition 4. The follow statements hold for all subsets $X \subseteq [k]$.

(i) Up to padding by zeros, the eigenvalues of $\tilde{\Phi}_X$, Φ_X and H_X are all identical.

$$\lambda(\tilde{\Phi}_X) \simeq \lambda(\Phi_X) \simeq \lambda(H_X). \quad (47)$$

(ii) The eigenvalues of Γ_X are equal to the eigenvalues of $\tilde{\Phi}_X$ plus the additional value of $1 - \text{Tr}(H_X)$.

$$\lambda(\Gamma_X) \simeq \lambda(\tilde{\Phi}_X) \oplus \{1 - \text{Tr}(H_X)\}. \quad (48)$$

(iii) The partial trace of $\Gamma_{[k]}$ over $[k] \setminus X$ is equal to Γ_X :

$$\text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) = \Gamma_X. \quad (49)$$

Proof. (i) From the definition of Φ_X , one observes that $\langle i_1 \dots i_k | \Phi_X | i_1 \dots i_k \rangle$ is only non-zero if there exists a unique non-zero $i_p \neq 0$ where $p \in X$ and thus

$$\text{Tr}(\Phi_X) = \sum_{p \in X} \sum_{j=1}^d \langle j | H_p | j \rangle = \sum_{p \in X} \text{Tr}(H_p) = \text{Tr}(H_X). \quad (50)$$

More generally, for any positive integer $\ell \in \mathbb{N}_{>0}$, one observes that $\text{Tr}(\Phi_X^\ell) = \text{Tr}(H_X^\ell)$. (**TODO: Prove this directly.**) As the eigenvalues of any matrix X are determined (up to padding by zeros) by the values of $\text{Tr}(X^\ell)$, we conclude that Φ_X and H_X have the same eigenvalues (up to padding by zeros) (**TODO: Is there a more direct way to prove this.**)

(ii) Note that $\tilde{\Phi}_X$ and its eigenvectors are supported in the subspace of $(\mathbb{C}^{d+1})^{\otimes |X|}$ spanned by $|i_1 \dots i_{|X|}\rangle$ for which exactly one if the i_p 's is non-zero. Therefore $|0 \dots 0\rangle$ is orthogonal to all eigenvectors of $\tilde{\Phi}_X$ and thus $|0 \dots 0\rangle$ is an eigenvector of Γ_X with eigenvalue $1 - \text{Tr}(H_X)$.

(iii) The partial trace of $\Phi_{[k]}$ over $[k] \setminus X$, when performed in the computational basis $|i_1 \dots i_k\rangle$, can be divided into three portions depending on the number of non-zero entries in $(i_p \mid p \in [k] \setminus X)$; first, is all indices i_p are zero, second, where exactly one i_p is non-zero, and third where more than one index i_p is non-zero. The respective contributions are as follows:

$$\text{Tr}_{[k] \setminus X}(\Phi_{[k]}) = \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + 0 \quad (51)$$

Recalling the definition of $\Gamma_{[k]}$,

$$\Gamma_{[k]} := \Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]} \quad (52)$$

One concludes

$$\begin{aligned} \text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) &= \text{Tr}_{[k] \setminus X}(\Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]}), \\ &= \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + (1 - \text{Tr}(H_{[k]}))Z_X, \\ &= \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \\ &= \Gamma_X. \end{aligned} \quad (53)$$

□

Corollary 10. *By Lemma 9, one has $\text{Tr}(\Gamma_X) = 1$. If the eigenvalues of H_X satisfy*

$$\text{Tr}(H_X) \leq 1, \quad (54)$$

then Γ_X is positive semidefinite, and thus Γ_X is a density matrix. In such a case, assuming $H_X \neq 0$ so that $\text{Tr}(H_X) \in (0, 1]$, the von Neumann entropy of Γ_X is related to the von Neumann entropy of $H_X/\text{Tr}(H_X)$ by the equation ²

$$S(\Gamma_X) = h(\text{Tr}(H_X)) + \text{Tr}(H_X) S\left(\frac{H_X}{\text{Tr}(H_X)}\right). \quad (55)$$

Moreover, if the eigenvalues of H_X satisfy the stronger condition

$$\lambda_1(H_X) + \text{Tr}(H_X) \leq 1, \quad (56)$$

then the maximum eigenvalue of Γ_X is $1 - \text{Tr}(H_X)$.

Proof. The only non-trivial claim involves the relation between the von Neumann entropies of Γ_X and $H_X/\text{Tr}(H_X)$, respectively given by

$$S(\Gamma_X) = -(1 - \text{Tr}(H_X)) \log(1 - \text{Tr}(H_X)) - \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X), \quad (57)$$

$$S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = -\frac{1}{\text{Tr}(H_X)} \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X) + \log(\text{Tr}(H_X)). \quad (58)$$

Multiplying the von Neumann entropy of $H_X/\text{Tr}(H_X)$ by $\text{Tr}(H_X)$ yields

$$\text{Tr}(H_X) S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = S(\Gamma_X) + \text{Tr}(H_X) \log(\text{Tr}(H_X)) + (1 - \text{Tr}(H_X)) \log(1 - \text{Tr}(H_X)), \quad (59)$$

which is the claimed result. $\square$

5 Discussion

²Here $h(p) = -p \log p - (1 - p) \log(1 - p)$ denotes the binary entropy of $p \in [0, 1]$.

Reference notes

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection to hyperbolic geometry quite explicitly.
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- An excellent expository overview of the big picture by Knutson [Knu00]
- The masters essay/thesis of Backens (TODO: check if the pdf Matthias sent to me is the final version) [Bac10]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]
- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- Klyachko [Kly00]
- Christandl [Chr08]
- Christandl, Şahinoğlu, and Walter [CSW18]
- Kirillov and Reshetikhin [KR90]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]

- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Boalch [Boa07]
- Taylor and Woodward [TW06]
- Henriques and Kamnitzer [HK05]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Potential application to inverse kinematics of 3-link arms in three dimensions? (Thanks Harold!)
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko

- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).
- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Horn's original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn's inequalities in characterizing the existence of invariant vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]

A Proof of Young recoupling upper bound

sec:young_recouple_bound_proof

The purpose of this section is to prove Theorem 8.

Proof. The proof follows by noting that the Young recoupling symbol can be written as the contraction of two smaller tensors across $V_\nu \otimes V_\gamma \otimes V_\mu \otimes V_\alpha$ (annotated below):

$$\text{Diagram (60)} \quad (60)$$

Applying the Cauchy-Schwarz inequality to $\left\| \begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} \right\|_{\text{HS}}^2$ (with respect to this contraction), and noting that

$$\text{Diagram (61)} \quad (61)$$

holds because the $Y_{\alpha\beta}^\mu$ and $Y_{\beta\gamma}^\nu$ maps are isometries, and moreover

$$\begin{array}{c}
 \begin{array}{c}
 C_{\alpha,\nu}^\lambda \\
 \downarrow \\
 \begin{array}{c}
 \text{---} V_\nu \text{---} \\
 \uparrow V_\lambda \\
 \text{---} V_\gamma \text{---} \\
 \downarrow V_\mu \\
 \text{---} V_\alpha \text{---}
 \end{array} \\
 \downarrow \\
 C_{\mu,\gamma}^\lambda
 \end{array}
 \begin{array}{c}
 \downarrow \\
 \text{---} V_\lambda \text{---} \\
 \downarrow \\
 C_{\alpha,\nu}^\lambda
 \end{array}
 \end{array}
 = \frac{\begin{array}{c} C_{\alpha,\nu}^\lambda \quad C_{\alpha,\nu}^\lambda \\ \text{---} V_\nu \quad V_\gamma \quad V_\mu \quad V_\alpha \\ \text{---} V_\lambda \end{array}}{C_{\mu,\gamma}^\lambda \quad C_{\mu,\gamma}^\lambda} \quad (62)$$

holds because of Schur's lemma (applied to V_λ) and that $Y_{\mu\gamma}^\lambda$ and $Y_{\alpha\nu}^\lambda$ are also isometries, yields the following upper bound:

$$(d_\alpha d_\beta d_\gamma)^2 \left\| \begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} \right\|_{\text{HS}}^2 \leq \left[(d_\alpha d_\beta d_\gamma)^2 \left(\frac{d_\nu d_\gamma d_\mu d_\alpha}{d_\lambda} \right)^2 c_{\alpha,\beta}^\mu c_{\beta,\gamma}^\nu c_{\mu,\gamma}^\lambda c_{\alpha,\nu}^\lambda \right]^{\frac{1}{2}}, \quad (63)$$

which is the statement of Theorem 8. □

References

- [AH11] Daiki Akimoto and Masahito Hayashi. “Discrimination of the change point in a quantum setting”. In: *Physical Review A* 83.5 (2011), p. 052328.
- [Ale+18] Anton Alekseev et al. “Concentration of symplectic volumes on Poisson homogeneous spaces”. In: *arXiv preprint arXiv:1808.06975* (2018).
- [Ale97] A Yu Alekseev. “On Poisson actions of compact Lie groups on symplectic manifolds”. In: *Journal of Differential Geometry* 45.2 (1997), pp. 241–256.
- [AM07] Anton Alekseev and Eckhard Meinrenken. “Ginzburg-Weinstein via Gelfand-Zeitlin”. In: *Journal of Differential Geometry* 76.1 (2007), pp. 1–34.
- [AMS20] Victor Alekseev, Andrey Morozov, and Alexey Sleptsov. “Multiplicity-free $U_q(\mathfrak{sl}N)$ 6-j symbols: relations, asymptotics, symmetries”. In: *Nuclear Physics B* 960 (2020), p. 115164.
- [AMW00] Anton Alekseev, Eckhard Meinrenken, and Christopher Woodward. “Group-valued equivariant localization”. In: *Inventiones mathematicae* 140.2 (2000), pp. 327–350.
- [AMW01] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Linearization of Poisson actions and singular values of matrix products”. In: *Annales de l’institut Fourier*. Vol. 51. 6. 2001, pp. 1691–1717.
- [AMW99] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Duistermaat-Heckman distributions for group valued moment maps”. In: *arXiv preprint math/9903087* (1999).
- [APS17a] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “A symplectic proof of the Horn inequalities”. In: *Advances in Mathematics* 318 (2017), pp. 711–736.
- [APS17b] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “The Horn problem and planar networks”. In: *Advances in Mathematics* 318 (2017), pp. 678–710.
- [ARS88] Robert Alicki, Sl/awomir Rudnicki, and Sl/awomir Sadowski. “Symmetry properties of product states for the system of N n -level atoms”. In: *Journal of mathematical physics* 29.5 (1988), pp. 1158–1162.
- [AW97] Sharad Agnihotri and Chris Woodward. “Eigenvalues of products of unitary matrices and quantum Schubert calculus”. In: *arXiv preprint alg-geom/9712013* (1997).
- [Bac10] Miriam Backens. “The 6j symbols and an extended version of Horn’s problem”. In: (2010).
- [Bar03] John W Barrett. “Geometrical measurements in three-dimensional quantum gravity”. In: *International Journal of Modern Physics A* 18.suppl02 (2003), pp. 97–113.
- [Bel08] Prakash Belkale. “Quantum generalization of the Horn conjecture”. In: *Journal of the American Mathematical Society* 21.2 (2008), pp. 365–408.
- [Bha01] Rajendra Bhatia. “[Linear algebra to quantum cohomology: the story of Alfred Horn’s inequalities](#)”. In: *The American Mathematical Monthly* 108.4 (2001), pp. 289–318.
- [BK16] Prakash Belkale and Shrawan Kumar. “The multiplicative eigenvalue problem and deformed quantum cohomology”. In: *Advances in Mathematics* 288 (2016), pp. 1309–1359.
- [Boa07] Philip P Boalch. “[Regge and Okamoto symmetries](#)”. In: *Communications in mathematical physics* 276.1 (2007), pp. 117–130.
- [BVW18] Nicole Berline, Michèle Vergne, and Michael Walter. “The Horn inequalities from a geometric point of view”. In: *L’Enseignement Mathématique* 63.3 (2018), pp. 403–470.
- [Chr08] Matthias Christandl. “A Quantum Information-Theoretic Proof of the Relation between Horn’s Problem and the Littlewood-Richardson Coefficients”. In: *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2008, pp. 120–128.
- [CMZ19] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “Revisiting Horn’s problem”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2019.9 (2019), p. 094018.
- [CMZ20] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “On Horn’s problem and its volume function”. In: *Communications in Mathematical Physics* 376.3 (2020), pp. 2409–2439.

- [Coq23] Robert Coquereaux. “About integer-valued variants of the theta and 6j symbols”. In: *Journal of Mathematical Physics* 64.3 (2023).
- [CP95] Vyjayanthi Chari and Andrew N Pressley. *A guide to quantum groups*. Cambridge university press, 1995.
- [CŞW18] Matthias Christandl, M Burak Şahinoğlu, and Michael Walter. “Recoupling coefficients and quantum entropies”. In: *Annales Henri Poincaré*. Vol. 19. 2. Springer. 2018, pp. 385–410.
- [CW06] Nicola Ciccoli and Pawel Witkowski. “From Poisson to quantum geometry”. In: *Lecture Notes, available at <http://toknotes.mimuw.edu.pl>* (2006).
- [CZ18] Robert Coquereaux and Jean-Bernard Zuber. “From orbital measures to Littlewood–Richardson coefficients and hive polytopes”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 339–386.
- [DH05] Sumit Daftuar and Patrick Hayden. “Quantum state transformations and the Schubert calculus”. In: *Annals of Physics* 315.1 (2005), pp. 80–122.
- [Esp12] Chiara Esposito. “On the classical and quantum momentum map”. In: (2012). 1203.4112: [arxiv](https://arxiv.org/abs/1203.4112).
- [Feg78] HD Fegan. “The heat equation on a compact Lie group”. In: *Transactions of the American Mathematical Society* 246 (1978), pp. 339–357.
- [FK00] Laurent Freidel and Kirill Krasnov. “Simple spin networks as Feynman graphs”. In: *Journal of Mathematical Physics* 41.4 (2000), pp. 1681–1690.
- [FKL10] Laurent Freidel, Kirill Krasnov, and Etera R Livine. “Holomorphic factorization for a quantum tetrahedron”. In: *Communications in Mathematical Physics* 297.1 (2010), pp. 45–93.
- [FL03] Laurent Freidel and David Louapre. “Asymptotics of 6j and 10j symbols”. In: *Classical and Quantum Gravity* 20.7 (2003), p. 1267.
- [FNR07] L Freidel, K Noui, and Ph Roche. “6J symbols duality relations”. In: *Journal of mathematical physics* 48.11 (2007).
- [FT14] Peter J Forrester and Colin J Thompson. “The Golden-Thompson inequality: Historical aspects and random matrix applications”. In: *Journal of Mathematical Physics* 55.2 (2014).
- [Ful00] William Fulton. “Eigenvalues, invariant factors, highest weights, and Schubert calculus”. In: *Bulletin of the American Mathematical Society* 37.3 (2000), pp. 209–249.
- [Hes20] Sarah Tanja Hess. “Examination of the connection between the Horn Problem and the Lax Conjecture”. In: (2020).
- [HHR16] Hal M Haggard, Muxin Han, and Aldo Riello. “Encoding curved tetrahedra in face holonomies: phase space of shapes from group-valued moment maps”. In: *Annales Henri Poincaré*. Vol. 17. Springer. 2016, pp. 2001–2048.
- [HK05] Andre Henriques and Joel Kamnitzer. *Crystals and coboundary categories*. 2005. arXiv: [math/0406478](https://arxiv.org/abs/math/0406478) [[math.QA](https://arxiv.org/abs/math/0406478)].
- [HK06] André Henriques and Joel Kamnitzer. “The octahedron recurrence and gl_n crystals”. In: *Advances in Mathematics* 206.1 (2006), pp. 211–249.
- [HM02] Masahito Hayashi and Keiji Matsumoto. “Quantum universal variable-length source coding”. In: *Physical Review A* 66.2 (2002), p. 022311.
- [Hor62] Alfred Horn. “Eigenvalues of sums of Hermitian matrices.” In: (1962).
- [Kly00] Alexander A Klyachko. “Random walks on symmetric spaces and inequalities for matrix spectra”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 37–59.
- [Knu00] Allen Knutson. “The symplectic and algebraic geometry of Horn’s problem”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 61–81.
- [KR90] AN Kirillov and N Yu Reshetikhin. “Representations of the Algebra Uq(sl(2)), q-Orthogonal Polynomials and Invariants of Links”. In: *New developments in the theory of knots* 11 (1990), p. 202.

- [KTW03] Allen Knutson, Terence Tao, and Christopher T Woodward. “A positive proof of the Littlewood-Richardson rule using the octahedron recurrence”. In: *arXiv preprint math/0306274* (2003).
- [KW01] Michael Keyl and Reinhard F Werner. “[Estimating the spectrum of a density operator](#)”. In: *Physical Review A* 64.5 (2001), p. 052311.
- [Lam05] McKenzie Lamb. “A proof of the Thompson conjecture using Poisson geometry”. In: (2005).
- [LE03] Jiang-Hua Lu and Sam Evens. “Thompson’s conjecture for real semi-simple Lie groups”. In: (2003). arXiv: [math/0310098](#).
- [McS21] Colin McSwiggen. “The Harish-Chandra integral: an introduction with examples”. In: *L’Enseignement Mathématique* 67.3 (2021), pp. 229–299.
- [Ric02] G Richardson. “The trigonometry of the tetrahedron”. In: *The Mathematical Gazette* 2.32 (1902), pp. 149–158.
- [Rob02] Justin Roberts. “Asymptotics and 6j-symbols”. In: *Geom. Topol. Monogr* 4 (2002), pp. 245–261.
- [Rob99] Justin Roberts. “[Classical 6j-symbols and the tetrahedron](#)”. In: *Geometry & Topology* 3.1 (1999), pp. 21–66.
- [RR86] David P Robbins and Howard Rumsey Jr. “Determinants and alternating sign matrices”. In: *Advances in Mathematics* 62.2 (1986), pp. 169–184.
- [SBT17] David Sutter, Mario Berta, and Marco Tomamichel. “[Multivariate trace inequalities](#)”. In: *Communications in Mathematical Physics* 352.1 (2017), pp. 37–58. arXiv: [1604.03023v3](#).
- [Spe05] David E Speyer. “Horn’s problem, Vinnikov curves, and the hive cone”. In: (2005).
- [TT73] RC Thompson and S Therianos. “The Eigenvalues and Singular Values of Matrix Sums and Products. VII (1)”. In: *Canadian Mathematical Bulletin* 16.4 (1973), pp. 561–569.
- [TW06] Yuka U Taylor and Christopher T Woodward. “6 j symbols for and non-Euclidean tetrahedra”. In: *Selecta Mathematica* 11.3-4 (2006), p. 539.
- [Zub18] Jean-Bernard Zuber. “Horn’s problem and Harish-Chandra’s integrals. Probability density functions”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 309–338.